

Is There a Puzzle in Underwater Mortgage Default?

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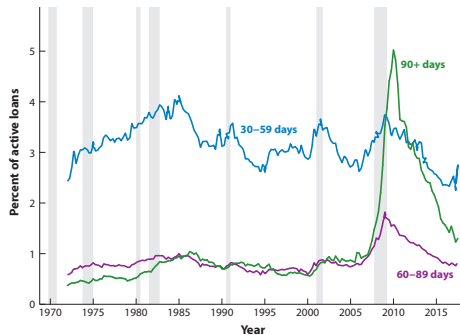
Disclaimer

- I am speaking today as a researcher and as a concerned citizen.
- Not as a representative of:
 - ▶ The Boston Fed,
 - ▶ or the Federal Reserve System.
- When I say “we”, I don’t mean:
 - ▶ Susan Collins and me, or...
 - ▶ Kevin Warsh and me.
- Everything I’m about to say could be wrong.
 - ▶ Economists can’t predict the future. They can’t even predict the past.



Motivation: the underwater default puzzle

- Mortgage defaults during the 2007–2010.
- Foote and Willen (2018). Theoretical models of default predicted **too many defaults**.
- To get realistic default rates, the literature added large nonpecuniary utility penalties on default:
 - ▶ Campbell & Cocco (2015): $\sim 25\%$ of lifetime consumption.
 - ▶ Laufer (2018): $\sim 29\%$ of permanent income.
 - ▶ Schelkle (2018): similar magnitudes.
- **This paper:** a realistic treatment of rents (and a richer choice set) **shrinks the required penalty fortyfold** — to 0.63% of lifetime consumption.



Mortgage delinquency rates, US (Foote & Willen 2018, Fig. 1)

Where we are going

1. Why rent should matter

Default compares two flows, not an option on prices — and rents did not fall with prices.

2. The model, the calibration, and the data

Default is two decisions. A penalty worth 0.63% of lifetime consumption fits both — and an IV agrees on the magnitude.

3. Questions and answers

Campbell–Cocco, the Ganong–Noel fact, where we fail, policy.

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The landlord's choice each period

The default decision reduces to:

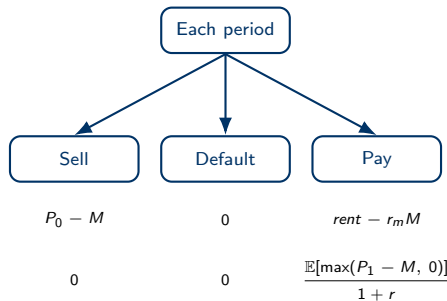
Pay if

$$\underbrace{rent - r_m M}_{\text{cash flow}_0} + \frac{1}{1+r} \underbrace{\mathbb{E}[\max(P_1 - M, 0)]}_{\text{cash flow}_1} > 0$$

or

$$\underbrace{\frac{\mathbb{E}[\max(P_1 - M, 0)]}{1+r}}_{\text{cost of default}} < \underbrace{r_m M - rent}_{\text{benefit of default}}$$

Otherwise default.

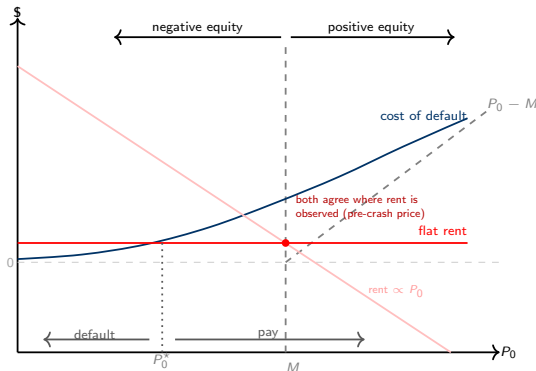


When does the landlord default?

Default iff

$$\underbrace{\frac{\mathbb{E}[\max(P_1 - M, 0)]}{1 + r}}_{\text{cost of default}} < \underbrace{r_m M - \text{rent}}_{\text{benefit of default}}$$

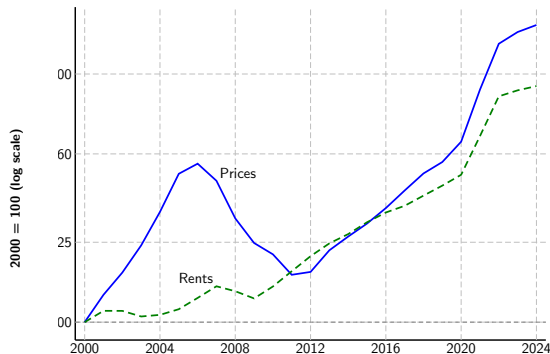
- **Problem:** a realistically calibrated model predicts too many defaults — P_0^* is close to M .
- **Literature's fix:** a nonpecuniary penalty ψ shifts the red line *down*, moving P_0^* left.
- **Our argument:** the line should not be that steep. Both rent assumptions must match the rent we observe, at the pre-crash price — so flattening the line pivots there, and P_0^* moves far left with no penalty.



- **Special case:** if $\text{rent} > r_m M$, you do *not* default regardless of LTV. Keep this in mind.
- Intuitively, if you have positive cash flow you never default.

Prices crashed. Rents did not.

- Standard calibrations assume rent tracks price.
 - ▶ price falls
 - ▶ rent falls
 - ▶ default more attractive
- But we had a big **boom and bust in house prices** (price line)
 - ▶ peak to trough, roughly a third
- **Rents did not fall much** — roughly constant in real terms (rent line).
 - ▶ default option much less valuable
- That is the whole intuition.
- Everything after this slide is establishing that it holds quantitatively.



1. Why rent should matter

Default compares two flows, not an option on prices — and rents did not fall with prices.

2. The model, the calibration, and the data

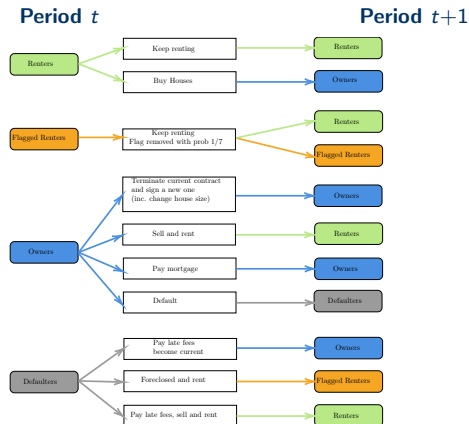
Default is two decisions. A penalty worth 0.63% of lifetime consumption fits both — and an IV agrees on the magnitude.

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Campbell–Cocco, the Ganong–Noel fact, where we fail, policy.

From two periods to a life cycle

- Example from above
 - ▶ One house
 - ▶ one date
 - ▶ no income
- A standard dynamic life-cycle model:
 - ▶ 58 years of age, savings, bequests
 - ▶ Stochastic income — expansion/recession mixture (Güvener et al. 2014)
 - ▶ Stochastic house prices, estimated on CBSA HPIs
 - ▶ Heterogeneous house sizes, smallest only for rent (Kaplan–Mitman–Violante 2020)
 - ▶ 30-year FRM: originate, refi, sell, default, cure (Campbell–Cocco 2015; Laufer 2018)
- Endogenous tenure choice.
- The decision tree is far more complex but intuition is the same.



What else makes default attractive

In the two-period model, the only short-run benefit of default was

$$\underbrace{r_m M - \text{rent}}_{\text{payment} - \text{rent}}.$$

In a real life cycle, default buys three more things:

- **A smaller house.** Not renting the same one back
 - ▶ After a permanent income shock you *want* less space
 - ▶ So default pays even where rent per square foot is high
- **A free year.** Stop paying, stay put
 - ▶ Housing at zero cost until anything is resolved
- **Ownership again.** Flag clears with probability 1/7 per year
 - ▶ A seven-year detour, not exile
 - ▶ “Never own again” is a penalty in disguise

Default got every advantage we could give it. Still not worth much.

Calibration

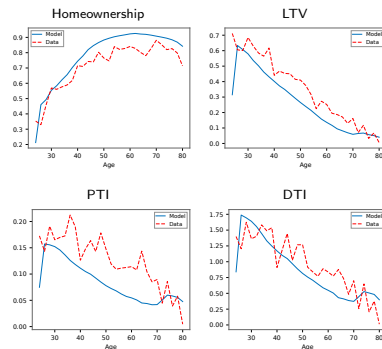
- **8 parameters, 8 targets**, searched jointly
 - ▶ 4 life-cycle profiles (2001 SCF/ACS)
 - ▶ 4 delinquency moments (CRISM)
- **Gauss–Newton** on normalized squared discrepancies
 - ▶ Converged iteration 9, objective **0.239**
- $\psi = 0.13$ in utility units
 - ▶ **0.63% of lifetime consumption** (CEV)
- **Model LTV measured as the data measure it**
 - ▶ Entry-year balance, log-error s.d. 0.14
- **No ownership premium**
 - ▶ Discipline is unit-level rent-to-price (Loewenstein 2023)

Calibrated parameters

		Value
β	Discount factor	0.92 [†]
ξ	Substitutability	1.98
η	Housing share	0.21
B_0	Bequest level	102
B_1	Bequest shifter	2.0
$h_{\min}$	Min. owned house	0.8 [†]
ψ	Default penalty	0.13
qT	Distress arrival	0.0011

[†] At a search-box edge; both bind toward *more* default, so the fit is not flattered.

Targets I: life-cycle profiles



Targets II: delinquency moments

	Model	Data	Fit
Delinquency by LTV, 15 bins	—	—	0.067
Above-water delinquency, %/yr	2.11	2.03	0.002
Underwater delinquency, %/yr	9.85	10.12	0.001
Above-water foreclosure delinquency	0.375	0.402	0.005

Default is two decisions, not one

- **Stage 1 — an owner stops paying**

- ▶ Four options, not two
- ▶ Default = *enter delinquency*
- ▶ Nothing terminal happens yet

- **Stage 2 — a defaulter resolves it**

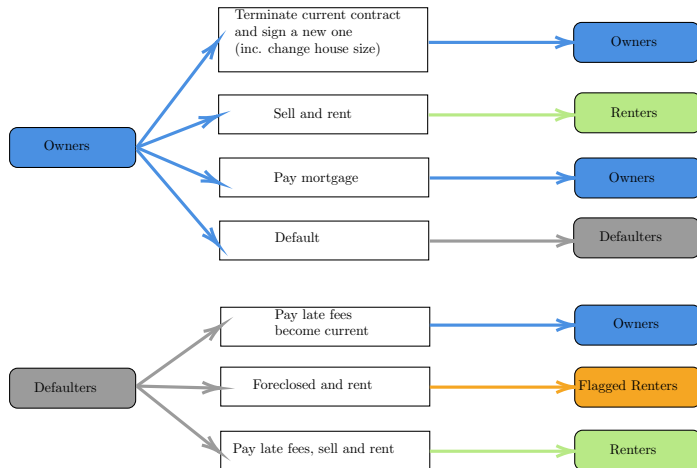
- ▶ Cure: pay arrears + 5% late fee
- ▶ Sell: pay off mortgage, -6% cost
- ▶ Foreclose

- **Equity is strongly related to delinquency**

- ▶ Who stops paying tracks equity closely
- ▶ But equity does not pay the arrears
- ▶ So it does not guarantee an *exit*

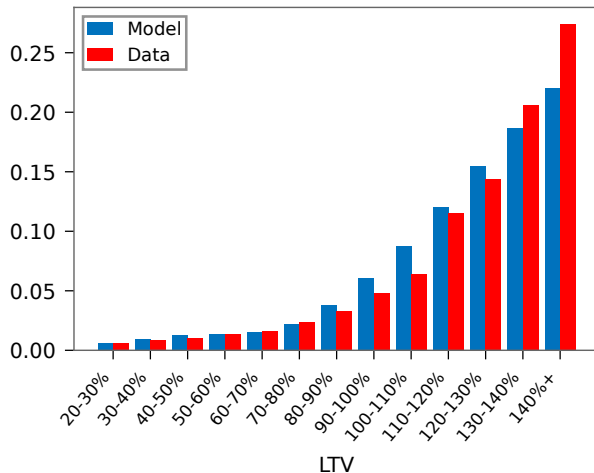
- **So we show the two stages separately**

- ▶ Who enters (next), then how it ends
- ▶ Above-water default lives in *both*



Default by LTV: model versus data

- **Stage 1: who enters delinquency**
 - ▶ First transition to 90+ days, CRISM
 - ▶ By LTV, 15 bins — a calibration target
- **The model matches it — level and slope**
 - ▶ At $LTV \geq 140$: 22% model, 27% data
 - ▶ Most deeply underwater borrowers keep paying
 - ▶ Option models put this near 100%
- **With a penalty of $\psi = 0.13$ — 0.63% of consumption**
 - ▶ Literature: 25% (CC), 29% (Laufer)
 - ▶ Fortyfold smaller — but not zero
 - ▶ $\psi = 0$ overshoots above water $4.7\times$
- **Above-water bars are positive — and should be**
 - ▶ Stage 1 is missing a payment, not losing the house
 - ▶ A liquidity event, available at any LTV
 - ▶ Nothing here is strategic



Model-implied LTVs carry measurement error with s.d. 14 percent. Bins start at 20–30%.

After default: cure, sell, or foreclose

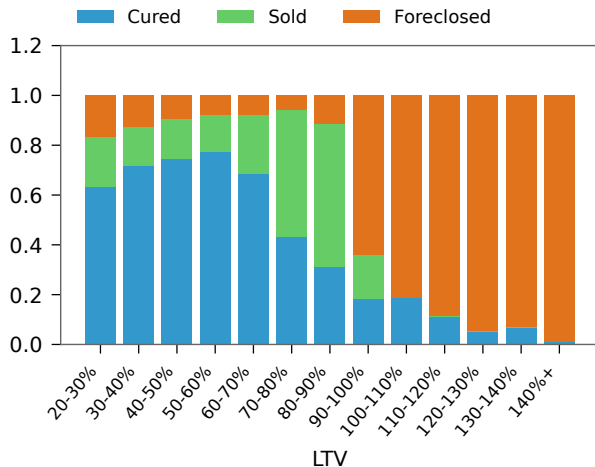
- **Stage 2: how delinquency ends**

- ▶ Above water, **curing dominates** (peak 0.78)
- ▶ Selling peaks *just below* water (0.50–0.58)
- ▶ Foreclosure rare until 90%, near-total after 100%

- **But some above-water borrowers still foreclose. Three reasons:**

- ▶ **Arrears.** Curing needs cash the borrower does not have; selling needs the full payoff plus 6%
- ▶ **Trigger shock** ($q_T = 0.0011$). Divorce, illness, a failed short sale
- ▶ **Measured equity.** Some were never above water (backup: the true-LTV counterfactual)

- **None of the three is a taste for default**



Resolution shares by true-LTV bin, no measurement error (paper Fig. 11c).

Where stage 2 fails: the shape, not the level

- **Foreclosure | delinquency is U-shaped, and wrong in both directions**

- ▶ Above water (30–70% LTV): 0.08–0.13 vs 0.32–0.44
- ▶ Deep underwater: 0.95 vs 0.66
- ▶ Entry we match — this is what happens *after*

- **Only the aggregate above-water rate is a target**

- ▶ The shape between the aggregates is undisciplined

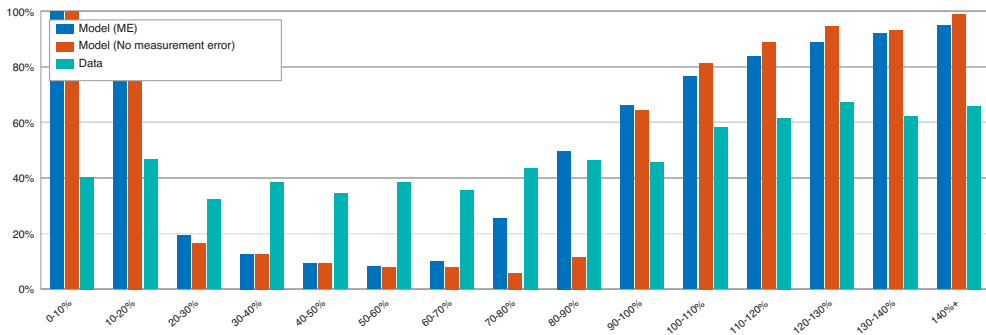
- **The life-cycle fit is imperfect too**

- ▶ Homeownership overpredicted at many ages
- ▶ Leverage and DTI discrepancies
- ▶ Matters: the LTV distribution cuts the later results

- **Neither failure runs through ψ**

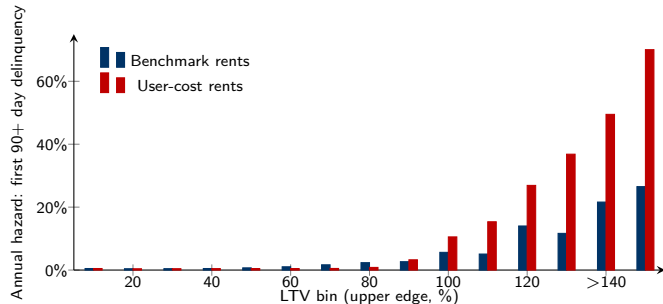
- ▶ ψ is fit to the *aggregate* above-water rate
- ▶ A re-calibration, not a re-identification

Foreclosure conditional on Delinquency by LTV



User-cost rents, our model

- **Same experiment, our model**
 - ▶ Every parameter at the calibrated point
 - ▶ Rent level → Campbell–Cocco user cost ($r + \delta + \tau - \mathbb{E} \Delta p$)
- **The puzzle is back, at a factor of 2.6**
 - ▶ Deep underwater: 26.5% → 70%
 - ▶ Data: 27%
- **And it is the slope, not a level shift**
 - ▶ Below 80% LTV default *falls*, 2.4% → 0.8%
 - ▶ Cheap rent makes *selling* attractive with equity
 - ▶ Above 90% it explodes — the option model, back
- **The rent process is doing the work, not the choice set**
 - ▶ Every generous option still switched on here



Transition off the calibrated stationary equilibrium: rent falls unexpectedly, ownership and leverage inherited — a Campbell–Cocco-style response, not a re-solved steady state. First year shown; years 2–5 within a few points. Aggregate delinquency 2.4% → 3.3–3.9%; homeownership 75% → 44%.

Empirical test: rent growth and foreclosure

- **Prediction.** Faster post-origination rent growth $\Rightarrow$ less foreclosure.
- **Sample.** CRISM first liens, originated pre-2008, first 90-day default 2010–2019. One observation per loan. $N = 2,511,128$; 25.4% foreclose.
- **Outcome.** $100 \times \mathbb{I}\{\text{foreclosed}\}$.
- **Regressor.** $\Delta \ln(\text{rent})$, CoStar CBSA effective rent/sq. ft., origination $\rightarrow$ default.
- **Controls.** Current LTV, $\ln(\text{payment})$, rate type, local wages, unemployment; year, origination-year, state FEs.
- **Why condition on default?** Everyone here has a liquidity problem. What varies is the cost of walking away.
- **IV (Gete–Reher 2018):** CBSA stress-tested-bank share. $IV > OLS$, so OLS is attenuated.

$$D_i = \beta_1 \Delta \ln(\text{rent}_{o,t(i)}) + \beta_2 LTV_i + \zeta X_i + \gamma_{t(i)} + \lambda_{o(i)} + \delta_i + \epsilon_i$$

	(1) OLS	(2) OLS	(3) IV
$\Delta \ln(\text{Rent})$	−0.86*** (0.06)	−0.65*** (0.06)	−2.86*** (0.72)
Current LTV	5.44*** (0.04)	5.40*** (0.04)	5.19*** (0.08)
$\ln(\text{Payment})$	−3.48*** (0.03)	−3.38*** (0.03)	−3.39*** (0.03)
Fixed rate	−3.36*** (0.06)	−3.49*** (0.06)	−3.46*** (0.06)
$\ln(\text{Avg. wages})$		−0.79*** (0.03)	−0.70*** (0.05)
Unemployment		−0.12** (0.06)	−0.58*** (0.16)
Adj. R^2	0.052	0.052	0.014
Local controls		Y	Y
Year, orig., state FE	Y	Y	Y
First-stage F			18,200

$N = 2,511,128$ in all columns.

Notes. Dependent variable $100 \times \mathbb{I}\{\text{foreclosure completed}\}$, mean 25.4. One observation per loan at its first 90-day default. All regressors standardized; $\Delta \ln(\text{rent})$ has s.d. 8.4 log points, so column (2) is −0.82 pp and column (3) −3.59 pp per 10% rent decline. *** $p < 0.01$, ** $p < 0.05$. Paper table *Probability of Foreclosure Conditional on 90-day Default*, cols. (1), (3), (4).

Model versus IV

- **One number, measured two ways**

- ▶ Extra foreclosure per delinquent borrower
- ▶ From a 10% cheaper outside option

- **Left bar: the model**

- ▶ Cut real rent 10%: 46.22 → 47.98%
- ▶ **1.76 pp**

- **Right bar: the IV**

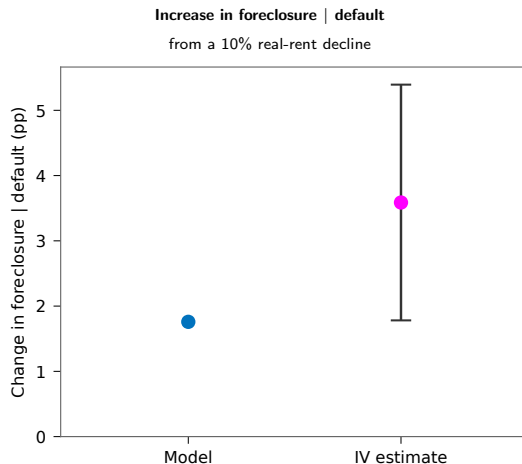
- ▶ Same 10% decline: **3.59 pp**, s.e. 0.90
- ▶ Whiskers ± 2 s.e.: [1.78, 5.39]

- **Same sign, same order of magnitude**

- ▶ Model at the low end of the band
- ▶ The structural number is the conservative one

- **And never a calibration target**

- ▶ A life-cycle model and a bank instrument agree
- ▶ A model that fits vs. the right mechanism



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Campbell–Cocco, the Ganong–Noel fact, where we fail, policy.

Ganong–Noel (2023)

- **Their diagnostic: the income drop at default**

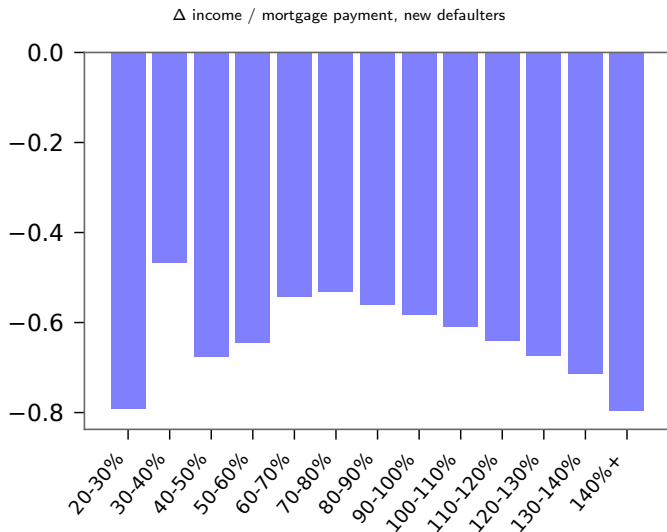
- ▶ Scaled by the monthly mortgage payment
- ▶ They find declines near one payment

- **Our model generates it**

- ▶ Income falls before default at every LTV
- ▶ -0.47 to -0.80 of a payment
- ▶ -0.64 above water, -0.82 below

- **Not a target**

- ▶ Nothing here was fit to an income profile



Campbell–Cocco (2015)

- **Not our model — theirs**

- ▶ Where the large penalty was estimated
- ▶ Fit to the income declines we just saw
- ▶ Vary two things: rent process, penalty

- **Their baseline: rents fall with prices**

- ▶ Needs $\psi \approx 25\%$ of lifetime consumption
- ▶ To match G&N's income declines

- **Fix real rent at 2001 levels**

- ▶ Same fit, *no* penalty
- ▶ Add the 25% on top: buys almost nothing

- **The rent assumption was the penalty**

- ▶ Not a distaste for default
- ▶ An unrealistic rent process, absorbed

Campbell–Cocco model: does it match G&N's income declines at default?

	No penalty	$\psi \approx 25\%$ CEV
Rents fall with p	no	yes
Constant real rent	yes	yes, adds little

Paper Appendix Figure A.1, Panels (a)–(d).

A large ψ also makes the model *insensitive* to flows. Cut rent 10% permanently in their high-penalty baseline and default does not budge: **0.19% → 0.19%** (Appendix Table A.1).

Policy: forbearance

- **One condition: get the payment under the rent**
 - ▶ The short-run benefit of default is $r_m M - \text{rent}$
 - ▶ Postponement sets the payment to zero
 - ▶ So the benefit is *negative* — nobody walks
- **No friction needed for this**
 - ▶ Even as an option: postponing is a dollar of cash flow saved
 - ▶ Capitalizing the balance raises the strike
 - ▶ But value falls *less* than one-for-one in the strike
 - ▶ Same conclusion as G&N, Palmer, Indarte — different mechanism
- **But small in steady state**
 - ▶ Default 2.35 → 2.16, foreclosure 1.09 → 1.06
 - ▶ Households anticipate it and **lever up**
- **The gain is in the transition**
 - ▶ As a surprise: foreclosure 1.09 → **0.46** in a year
- **Insurance against a shock, not a lower default rate**

Default rate by equity position

paper table *Default and Foreclosure Outcomes: Abovewater vs. Underwater*, Panels A–C

	Above water	Underwater
Data (CRISM)	2.03	10.12
Baseline model	2.11	9.85
With forbearance	1.87	7.05

Aggregate effect

paper table *Default and Foreclosure Outcomes: Mortgage Forbearance*

	Default	Foreclosure
Baseline	2.35	1.09
Stationary eq. with FB	2.16	1.06
Year of introduction	1.28	0.98
One year after	1.48	0.46

Forbearance: one-year payment postponement for mortgagors with PTI > 31%. Rows 3–4 are the transition after an unanticipated introduction.

Takeaways

- **Rent should matter, intuitively.** Default is a comparison of flow costs, not an option on house prices. Rents did not fall with prices, so the “puzzle” is an artifact of assuming they did.
- **Rent matters quantitatively, in the model.** With realistic rents and a realistic choice set — downsizing, post-default rebuying, a free year of housing — the penalty needed is **0.63%** of lifetime consumption, against 25% (Campbell–Cocco) and 29% (Laufer).
- **And by about the same amount in the data.** The flow-cost elasticity is not a calibration target, yet the model’s **1.76 pp** sits at the low edge of the two-s.e. band of an IV built from bank stress tests (**3.59 pp**, band [1.78, 5.39]).
- *Policy:* cash-flow relief is the effective lever, but forbearance is insurance against a shock, not a permanent cut in default — anticipating households lever up.
- *A caution for empirical work:* the Ganong–Noel strategic share is not identified without an assumption about the above-water comparison group. Inside our model we move it from 39% to zero while underwater default behavior is completely unchanged.

So why keep a penalty at all?

- Realistic rents shrink the penalty; they do not eliminate it. Ours is $\psi = 0.13$, **worth 0.63% of lifetime consumption** — fortyfold below the literature, but not zero.
- Could a restricted choice set do the rest? Limit foreclosed households to the smallest rental unit and default falls further (9.85% $\rightarrow$ 4.92% underwater).
- **But take the penalty away entirely and the model breaks** — default overshoots the data badly in *both* groups, 9.55% above water against 2.03% in CRISM.
- **Reading:** the puzzle is not that the penalty is implausibly large. It is that the earlier models had to make it large.

Default rate by equity position

paper table *Default and Foreclosure Outcomes: Abovewater vs. Underwater*

%	Above water	Underwater
Data (CRISM)	2.03	10.12
Our model, $\psi = 0.13$	2.11	9.85
Limited rental after fcl.	1.71	4.92
... and <i>no</i> penalty	9.55	19.22

Panels A, B, D, E. "Limited rental" restricts foreclosed households to the smallest rental unit.

On request: does this travel to Canada?

- Our mechanism is about **rents, the choice set, and liquidity** — none of which is US-specific. But the institutions differ, and they push in identifiable directions.
- **Recourse** raises the effective cost of default without any nonpecuniary penalty. Less need for ψ , not more.
- **Five-year terms with renewal** put a scheduled liquidity shock in every borrower's path. Our liquidity channel should be *stronger*, and predictable in timing.
- **CMHC insurance at high LTV** changes who bears the loss, so the lender-NPV calculus behind forbearance is a different calculation — but the borrower's problem is the same one.

Feature	US	Canada
Rate term	30-yr fixed	5-yr, renewed
Recourse	Varies by state	Mostly recourse
High-LTV ins.	FHA / PMI	CMHC
Prepayment	Free	Penalized
Rent dynamics	Loewenstein (2023)	?

The question for this room — the last row. We do not know whether Canadian rents track Canadian prices the way the models assume. If they do, the puzzle here is real and our resolution does not travel. If they do not, it should. Which of these institutions makes measured above-water default larger or smaller than ours?

On request: the Ganong–Noel strategic share

- Every default here is a double trigger
 - ▶ More equity would reduce it (0.4% → 26.5% across LTV)
 - ▶ More income would reduce it, at every LTV
 - ▶ **No equity-only group, no income-only group**
- So their statistic has no clean target here
 - ▶ It asks for the equity-*only* share
 - ▶ Using above-water defaulters as the pure-liquidity benchmark
- Note the direction
 - ▶ Underwater defaulters have the *larger* income drops
 - ▶ Profile deepens monotonically past 70% LTV
- And it is fragile
 - ▶ Start where it returns something (row 2 — baseline gives zero)
 - ▶ Shock the above-water *group only*: 39% → 0
 - ▶ Underwater behavior unchanged

G&N's income comparison

$$\hat{\theta}^{GN} = \max\{0, 1 - (\mu_U - \mu_0)/(\mu_A - \mu_0)\}, \mu_0 = 0.475$$

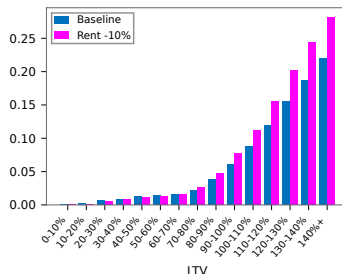
Scenario	Δy / payment		G&N-implied
	AW	UW	UW strategic share
Baseline	−0.636	−0.820	0.0%
+ UW default shock	−0.636	−0.201	39.1%
+ AW default shock too	−0.201	−0.201	0.0%

AW/UW split on true LTV at 100%, ages 24–79. Share floored at zero. Applies G&N's formula to annual model moments, not their monthly estimator. Paper table *Abovewater Default Selection and Implication for G&N*.

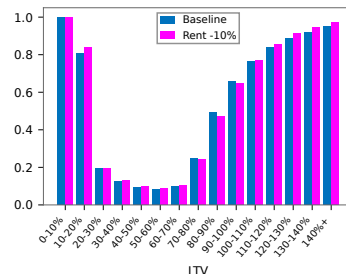
Underwater behavior is identical in rows 2 and 3. The share moves 39% → 0 anyway — it is a statement about the comparison group.

Counterfactual: a permanent 10% decline in real rent

- The cleanest test of the mechanism *inside* the model: make the outside option 10% cheaper, change nothing else. That is about the largest real-rent decline borrowers actually saw in 2007–2012 — though ex post that one was not permanent.
- Panel (a):** default rises **5.66 pp** at LTV 130–140% and **6.16 pp** above 140%, falling with LTV but positive all the way down — 1.64 pp at 90–100%, 0.45 at 70–80%.
- Panel (b):** foreclosure | delinquency rises **1.9–2.5 pp above 110% LTV** but *falls* in the intermediate bins (–2.1 pp at 80–90%) — composition of the delinquent pool, not a sign flip in behavior.
- Negative equity is not what creates the incentive — the **flow cost of the alternative** is. Hence a **triple trigger**: equity, income, and flow costs.



(a) Default rate by LTV

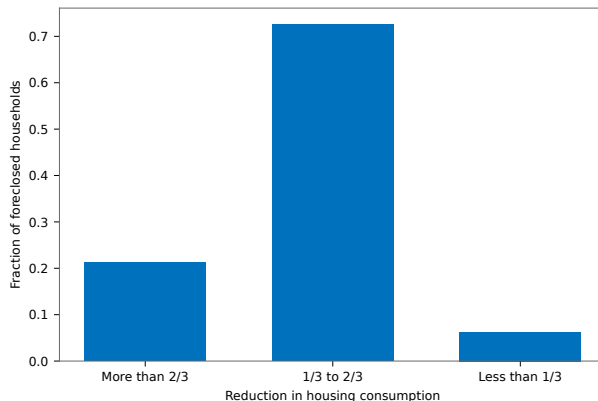


(b) Foreclosure rate | default

Model-implied LTVs carry measurement error with s.d. 14 percent.

The outside option: a smaller house

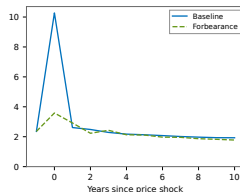
- A recurring objection: “surely renting the same house costs about what owning it costs.” But the relevant counterfactual is not renting the same house.
- Households that default have had an **income shock**, so their demand for space falls. Downsizing is part of the default payoff, and our choice set lets them do it.
- In the model, **72.5%** of foreclosed households end up in a unit one-third to two-thirds of their old size and **21.3%** go below one-third; only **6.3%** lose less than a third. The average is **52.8%** of the previously owned home.
- This is why the flow saving from walking away is real — and why we need only a small ψ to match the data.



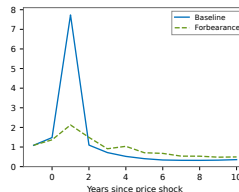
Ratio of post-foreclosure rented size to previously owned size.

Forbearance under a house-price crash

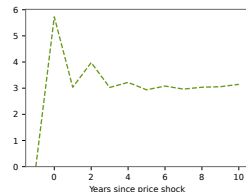
- The steady-state numbers assume households *anticipate* forbearance at origination and lever up accordingly — a conservative benchmark.
- The financial-stability question is different: start in the no-forbearance steady state, crash prices to the lowest state, and introduce forbearance at the same moment.
- **Default and foreclosure rise far less sharply** under the forbearance regime (Panels a, b).
- **Take-up peaks at 8%** and settles near 5% — a small, self-limiting program.
- Costless to a risk-neutral lender; *valuable* to one whose marginal utility is high in a crash.



(a) Delinquency



(b) Foreclosure



(c) Forbearance take-up

Does the answer depend on when you ask?

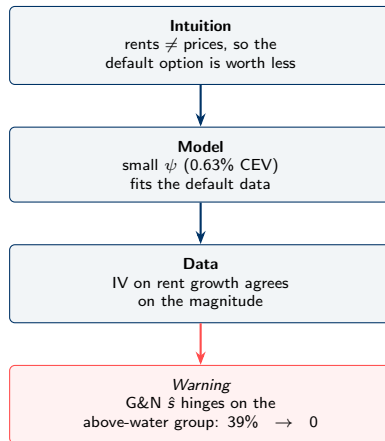
- We calibrate to **2001** rent-to-price ratios. That is deliberate: 2001 ratios are close to 2009 ratios, so the calibration sits on the crash period, where underwater default was salient.
- But rent-to-price has since fallen **about 30%** relative to 2001, with wide dispersion across markets. Owning is now expensive relative to renting in a way it was not.
- Mechanically, a lower rent-to-price ratio makes the renting alternative look *better* — so our channel weakens and the strategic incentive should be larger.
- **Whether it actually is depends on why people own at a low rent-to-price ratio.**

Two readings, opposite implications:

- If they own for an **ownership premium** or to hedge rent growth risk (Sinai–Souleles), the reason to keep paying is intact and strategic default stays moderated.
- If they own to **speculate on prices**, and the crash reset those expectations, then they really do have a strategic incentive when underwater.
- A small speculative group may be exactly why our baseline income gradient is steeper than G&N's data.
- A 30% lower rent-to-price paired with a 30% ownership premium gives *the same* household choices as our calibration.

Three things to take away

- **1. Rent should matter — intuitively**
 - ▶ Default compares two flows, not an option on prices
 - ▶ Rents did not fall with prices
 - ▶ So walking away was never as cheap as assumed
- **2. Rent matters quantitatively — in the model**
 - ▶ Required penalty: **0.63%** of lifetime consumption
 - ▶ Against 25% (Campbell–Cocco), 29% (Laufer)
- **3. And by about the same amount in the data**
 - ▶ Flow-cost elasticity is *not* a calibration target
 - ▶ Model 1.76 pp, at the low edge of the IV band
- *And on Ganong–Noel: we match their fact*
 - ▶ Every default here is a *double trigger*
 - ▶ So the equity-only share has no clean target
 - ▶ We move it 39% $\rightarrow$ 0, underwater behavior fixed



Which ingredient does the work? (within our model)

- We now have the rent row in *our* model — user-cost rents put the puzzle back. What we do not have is how much the *choice set* contributes on its own.
- The exercise: switch off one ingredient, hold the other seven parameters at their calibrated values, and see what happens to the moment ψ was fit to.
- **Rent realism alone is enough to break the fit.** What we cannot yet say is whether the choice set — downsizing, post-foreclosure rebuy, the free year — would break it too.

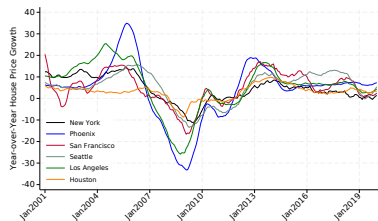
Switch off . . .	Deep-UW hazard	vs data 27%
nothing (baseline)	26.5%	matches
rent realism ($r \downarrow$ with p)	70.0%	over 2.6×
<i>And on the penalty itself:</i>		
ψ entirely, options intact	AW default 9.55% vs 2.03%	

What we can defend: rent realism alone brings the puzzle back, with every generous option still switched on — so the rent process is sufficient.

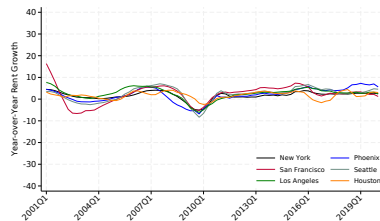
What we cannot yet say: how much the choice set contributes on its own. Turning off downsizing, the rebuy, the free year or k_s one at a time and re-fitting ψ needs four solves we do not have. The attribution is one-sided by construction, and I would rather say that than guess.

Backup: the same picture city by city

- The headline figure is a national index. The obvious worry is that it averages away the bust markets — Phoenix, Las Vegas, the Central Valley — where the underwater borrowers actually were.
- It does not. Rents fell in those cities too, but by **roughly an order of magnitude less than prices**: house-price growth troughs near -30% year over year, rent growth near -5 to -10% .
- So the flow comparison moves the way the national picture says it does, in exactly the places where the puzzle is measured.



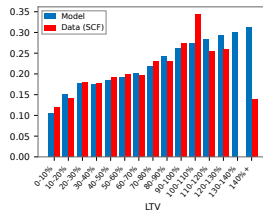
(a) House-price growth, CoreLogic HPI



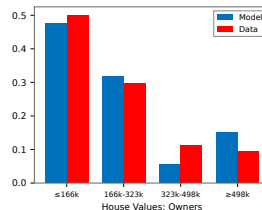
(b) Effective market rent growth, CoStar

Backup: fit in *nontargeted* moments

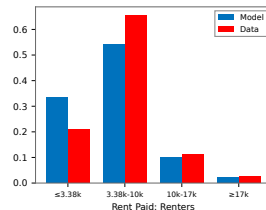
- Nothing here was targeted. The comparison is to the **2001 SCF**.
- Payment-to-income by LTV, the house-value distribution among owners, and the rent distribution among renters all line up.
- This matters for the argument: the flow-cost channel runs through payments and rents, so getting those distributions right is what makes the small ψ credible rather than a fitting artifact.



(a) PTI by LTV



(b) House values, owners



(c) Annual rents, renters

Backup: where we sit in the literature

Large penalties, structural:

- Campbell & Cocco (2015) — $\sim 25\%$ of lifetime consumption.
- Laufer (2018) — $\sim 29\%$ of permanent income.
- Schelkle (2018) — similar magnitudes.

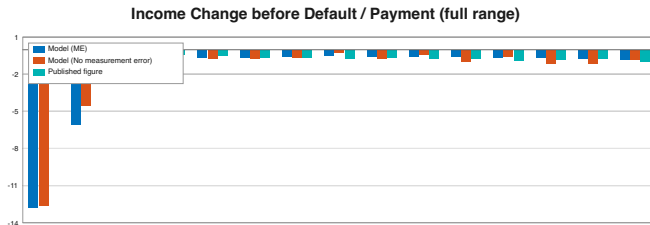
Liquidity and modest costs, empirical:

- Ganong & Noel — payment vs principal reduction.
- Palmer — IV hazard on house-price exposure.
- Indarte — liquidity vs strategic motives.
- Low; Brueckner et al. — default costs and choice sets.

- Our contribution is not a new estimate of the penalty. It is that **the penalty was never large** once the rent process is right.
- We land on the empirical side of the literature from a fully structural model — which is why the G&N result matters: we can audit their identification inside a world we control.

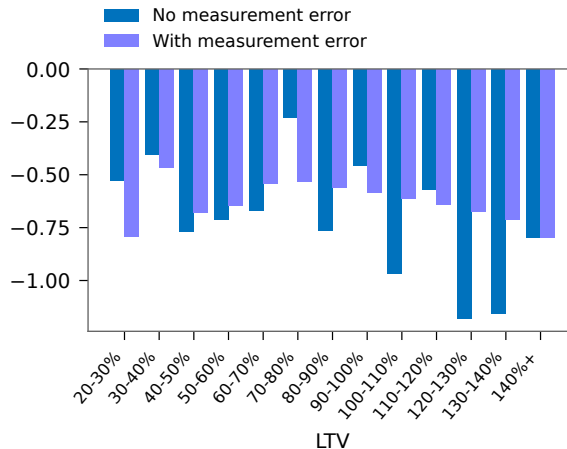
Backup: Δ income at default, untruncated

- The published figure truncates at 30% LTV. Below that the ratio explodes: -12.8 and -6.1 in the 0–10% and 10–20% bins.
- Mechanically: these households have collapsed income against a tiny remaining payment, so the denominator vanishes.
- They are under half a percent of defaulters — but they are why the published figure starts at 30%.



Backup: Δ income at default, true vs measured LTV

- Same object on raw true LTV rather than operator-weighted bins (paper Fig. 11b).
- On true LTV the profile is **jagged and deeper** — past -1 in the 110–130% bins — because the cells are thin.
- The operator compresses it into a narrow -0.4 to -0.8 band. It **smooths, it does not shift**: the average is the same.
- So the flat income-change profile we show in the body is not an artifact of the smoothing — but it is smoother than the truth.



Average one-year income-change/payment ratio among new defaulters, ages 24–79.

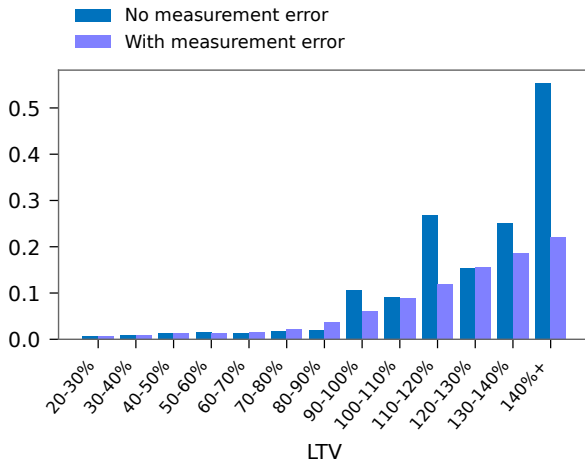
Backup: fit at the calibrated point

Moment	Model	Target	Fit term
Homeownership by age	—	—	0.011
LTV by age	—	—	0.041
DTI by age	—	—	0.089
PTI by age	—	—	0.024
Delinquency by LTV (15 bins)	—	—	0.067
Above-water delinquency /yr	0.0211	0.0203	0.002
Underwater delinquency /yr	0.0985	0.1012	0.001
AW foreclosure delinq.	0.375	0.402	0.005
Total			0.239

- Joint eight-parameter search beats the earlier two-parameter winner ($\psi = 0.14$, $q = 0.0018$; total 0.267).
- The gain is in the delinquency block: 0.100 $\rightarrow$ 0.074, mostly the bin profile.
- It trades a little of the AW foreclosure aggregate for the AW delinquency level.
- DTI by age (0.089) is the worst-fitting block — the mid-life leverage plateau again.

On request: how much does measurement error do?

- **Observed LTV is noisy**
 - ▶ Automated valuations: s.d. $\sigma \approx 14\%$ (Bogin et al.)
 - ▶ So some “above water” borrowers are not
- **So we compare data to a measured model object**
 - ▶ True-LTV hazards, reweighted by a Berkson operator
 - ▶ Bin weights from the measured-LTV histogram
- **This is load-bearing, and we say so**
 - ▶ True LTV, deep underwater: **0.55** vs 0.27 — over $2\times$
 - ▶ Measured: **0.22**. That is the 22-vs-27
- **Two caveats**
 - ▶ σ imported, never calibrated — and it *shrinks* ψ
 - ▶ Not an exact Berkson joint distribution
 - ▶ So we report true LTV throughout
- **And not the whole story: below 90% the curves coincide**



Default rate by LTV, same policies and population, no recalibration (paper Fig. 11a).